Project Proposal - Draft

Title: SpeculatorGate - A Reproducible Promotion Gate for Speculative Decoding Draft Models
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Project Type: Work-Based Project
Project Category: AI/ML Development, DevOps, Cloud Computing, CI/CD Pipeline

Problem Statement

Speculative decoding accelerates LLM inference by pairing a small drafter model with a verifier, but drafter promotion remains entirely manual. Hidden-state extraction, training sweeps, and performance validation are orchestrated ad hoc, with no reproducible record of which dataset revision, extraction configuration, or hyper-parameters produced a given drafter. Teams lack an automatic quality gate to prevent a slow or misaligned drafter from deployment, and on shared clusters repeated extraction jobs waste expensive GPU capacity.

Project Goal

This project aims to design, build, and evaluate a reproducible promotion gate for speculative decoding draft models. The system will support deployment decisions by providing statistical validation, token-level output consistency checks, workload robustness testing, and auditable decision lineage. The project evaluates whether SpeculatorGate can make reliable promotion decisions despite measurement noise, workload variability, and non-deterministic inference behaviour.

Background

While existing tools solve fragments, the Kubeflow SDK's SpeculativeDecodingTrainer automates training; MLflow tracks runs; Kubeflow Pipelines orchestrates steps. An evaluation component for speculative decoding will report speedup and acceptance metrics. SpeculatorGate adds a reproducible promotion framework to existing speculative decoding tooling based on statistical validation, token-level output consistency checks, and workload robustness testing. At the time of writing, the author is not aware of an open-source workflow that integrates statistical validation, output consistency checking, workload robustness testing, and auditable promotion decisions for speculative decoding draft models.

The intended users of SpeculatorGate are ML engineers, MLOps engineers, and platform teams responsible for training, evaluating, and deploying speculative decoding draft models.

Proposed Solution

SpeculatorGate is a correctness-aware promotion gate that consumes evaluation results and decides when a drafter is fit for deployment. Rather than rebuilding benchmarking, the project builds the decision framework:

- **Statistical confidence:** Promotion decisions are based on repeated baseline and speculative-decoding evaluations using confidence bounds on observed speedups, reducing the impact of measurement noise.
- **Output consistency:** Generated token sequences are compared between baseline and speculative executions. The gate records token divergence and applies configurable token-level acceptance policies.
- **Workload robustness:** Candidates are evaluated across representative workloads, including varying request rates, concurrency levels, and prompt datasets.
- **Full lineage:** Every decision records the dataset revision, extraction configuration, model architecture, hyper-parameters, metrics, and hardware used.

A draft model is promoted only when it demonstrates reproducible latency improvements over verifier-alone inference while preserving output quality. Promotion decisions are based on objective evaluation results rather than manual judgement.

Technologies

Python, PyTorch, vLLM, Kubeflow SDK (Trainer, Pipelines), Red Hat OpenShift AI (Data Science Pipelines) as the primary hosted platform subject to availability, local Kubernetes (kind) for development and CI fall-back when cluster access is not provisioned, MLflow (tracking/registry), YAML manifests, Git/GitHub, pytest, Trello, Harvard Referencing via Zotero. Evaluation will be performed using open-source benchmark prompt datasets and generated model outputs rather than personal or sensitive data.

Methodology

Development will follow an Agile approach with six two-week sprints, weekly supervisor reviews, and progress tracked using Trello. Work will be delivered incrementally, beginning with the promotion-gate core, followed by pipeline integration, empirical evaluation, and final artefacts.

Summary

SpeculatorGate delivers and evaluates a promotion gate for speculative decoding draft models, enabling deployment decisions to be based on reproducible evidence rather than manual judgement. By combining statistical validation, token-level output consistency checks, and testing across representative workloads, the framework provides clear criteria for promotion, rejection, or further investigation. Evaluation focuses on how effectively the implemented system supports promotion decisions under measurement noise, workload variation, and non-deterministic inference conditions.